FABRICATION OF POWER GENERATING SHOCK ABSORBER SYSTEM

A Project report

Submitted in the partial fulfilment of the

Requirement' s for the award of

BACHELOR OF TECHNOLOGY

IN

DEPARTMENT OF MECHANICAL ENGINEERING

Submitted

by

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CERTIFICATE

This is to certify that the project entitled

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In partial fulfilment of the requirements for the award of Bachelor of Technology in Department of Mechanical Engineering is record of Bonafide work carried out under guidance and supervision during the academic year 2024 2025. The results embodied in this project report have not been submitted in any other university or institute for the award of any degree or diploma

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DECLARATION

We hereby declare that the minor project work entitled "**FABRICATION OF POWER GENERATING SHOCK ABSORBER SYSTEM"** is a Bonafide work carried out by **Batch-02** during the academic year 2024-25 is submitted to Jawaharlal Nehru Technological University Hyderabad -University college of Engineering Sultanpur is a record of an original work done by us under the guidance of **A. MANOHAR**, Assistant Professor © Mechanical Engineering, JNTUH University College of Engineering Sultanpur. This project work is submitted in the partial fulfilment of the requirement for the award of the degree of bachelor of technology. The results embodied in this project have not been submitted to any other institute for the award of any degree/diploma/fellowship and similar project if any.

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ABSTRACT

This project deals with Fabrication of Regenerative shock absorber. A smart solution has emerged in which the wasted energy in a vehicle's shock absorber is converted to an alternative energy for the cars themselves and this is called an energy regenerative shock absorber.

The regenerative shock absorber has become more attractive to several researchers and industries in the past decade. Due to the roughness of the road surface, vibration has been provided between the road surface and car body when drives on irregular road surface. The function of regenerative shock absorber is to recover the vibration energy which can be dissipated in the form of heat as waste. This paper first introduces the existing research and significance of regenerative shock absorbers and reviews the potential of automotive vibration energy recovery techniques and then emphatically classifies and the general classifications of regenerative shock absorber. Finally, it analyses the modelling approaches of shock absorbers, actuators and dampers. With the increasingly prominent energy issues, regenerative shock absorber becomes one of promising trends of vehicles.

LITERATURE REVIEW

Peng Zheng et al. did extensive research has been performed regarding Review on the Research of Regenerative Shock Absorber in 2021. The regenerative shock absorber has become more attractive to several researchers and industries in the past decade. Due to the roughness of the road surface, vibration has been provided between the road surface and car body when drives on irregular road surface. The function of regenerative shock absorber is to recover the vibration energy which can be dissipated in the form of heat as waste. This paper first introduces the existing research and significance of regenerative shock absorbers and reviews the potential of automotive vibration energy recovery techniques, and then emphatically classifies and summarizes the general classifications of regenerative shock absorber. Finally, it analyses the modelling approaches of shock absorbers, actuators and dampers. With the increasingly prominent energy issues, regenerative shock absorber becomes one of promising trends of vehicles.

Seema Tiwari et al. did research on the regenerative Shock Absorber in 2020. Abstract Sustainable power source advancements, especially in electric vehicles (EVs), have gotten noteworthy consideration in recent years. Because of the anomalies out and about surfaces our vehicles experience shocks. The customary suspension framework in our vehicles lessen these shocks and convert this mechanical vibrational energy into heat. In this procedure a lot of fuel energy is squandered which can be recouped utilizing a regenerative suspension framework. A regenerative suspension framework can successfully retain these vibrations just as lessen the measure of energy lost to the encompassing. This paper audits the current research on the regenerative shock absorbers. It initially examines about the energy dispersal from the vehicles and afterward the capability of recouping this scattered energy utilizing a regenerative shock absorber. It additionally audits the different innovative work done on the regenerative shock absorber. Three drive modes of the regenerative shock absorber systems, namely the direct drive mode, the indirect drive mode and hybrid drive mode are reviewed for their readiness to be implemented. Different types of parameterized road excitations have been applied to vehicle suspension systems to investigate the performance of the regenerative shock absorbers including that of the nonlinear regenerative shock absorber. The research gaps for comparison of the different drive modes and the nonlinearity analysis of the regenerative shock absorbers are identified and, the corresponding research questions have been proposed for future work.

Huanyu Zhao et al. conducted research on research of a Vehicle Energy-Regenerative Suspension System in 2019. Vehicles are developing in the direction of energy-saving and electrification. suspension has been widely developed in the field of vehicles as a key component. Traditional hydraulic energy-supply suspensions dissipate vibration energy as waste heat to suppress vibration. This part of the energy is mainly generated by the vehicle engine. In order to effectively utilize the energy of this part, the energy-regenerative suspension with energy recovery converts the vibrational energy into electrical energy as the vehicle's energy supply equipment. This article reviews the hydraulically powered suspension of vehicles with energy recovery. The importance of such suspension in vehicle energy recovery is analysed. The main categories of energy-regenerative suspension are illustrated from different energy recovery methods, and the research status of hydraulic energy-regenerative

suspension is comprehensively analysed. Important factors that affect the shock-absorbing and regenerative characteristics of the suspension system are studied.

Zhan Wen Wang et al. did survey research on a high-efficiency regenerative shock absorber considering twin ball screws transmissions for application in range-extended electric vehicles in 2020. Renewable energy technologies, particularly in electric vehicles (EVs), have received significant attention in recent years. The wasted energy in a vehicle's shock absorber can be converted into an alternative energy source by regenerative shock absorbers. In this paper, a high-efficiency regenerative shock absorber considering twin ball screws transmissions is proposed for application in range-extended electric vehicles. The proposed regenerative shock absorber can convert vibrational kinetic energy, which is traditionally dissipated as heat in suspension systems, into electricity.

The proposed system is divided into four modules: suspension vibration input module, transmission module, generator module and power storage module

to the suspension vibration input module. The reciprocating vibrations are converted into unidirectional rotation of the generator by a pair of ball screws, gears, and two overrun clutches in the transmission module.

Ganesh M. did extensive research on a Review on the Regenerative Shock Absorber Systems of Four-wheeler Vehicles in2018. In this paper, the current technologies of the regenerative shock absorber systems have been categorized and evaluated. Three drive modes of the regenerative shock absorber systems, namely the direct drive mode, the indirect drive mode and hybrid drive mode are reviewed for their readiness to be implemented. The damping performances of the three different modes are listed and compared. the vehicle suspension systems to investigate the performance of the regenerative shock absorbers on different types of roads have been recorded.

Dr. S. Chakradhar Goud et al. did extensive research on analysis regenerative shock absorber in 2017. The vehicles normally release energy in the form of vibration which will be dissipated in heat format. This dissipated energy is usually considered as wastage.

In order to overcome this unwanted release of energy as well as to enhance the performance of vibration reduction, a technique called regenerative suspensions is motivating lot of researchers. The quantity of energy released additionally the capability of regenerating energy is provided in this present work. The research carried out in designing this regenerative suspension is reviewed in this paper.

CONTENTS

Certificate…….. II

Acknowledgement……. III

Declaration……. IV

Abstract……... V

Literature Review…… VI, VII

Contents……... VIII

Index……. IX

List of figures…… X

INDEX

S.NO	CONTENTS	PAGE NO
1	**Chapter-1**	
	1.1 Introduction	1
2	**Chapter-2**	
	2.1 Shock absorber	3
	2.2 Vehicle suspension	5
	2.3 Power generating shock absorber	6
	2.4 Working	7
	2.5 Components	7
3	**Chapter-3**	
	3.1 Fabrication process	15
4	**Chapter-4**	
	4.1 Advantages	21
	4.2 Disadvantages	22
	4.3 Applications	23
	4.4 Advancements	23
5	**Chapter-5**	
	5.1 Calculations	24
6	**Conclusion**	25
7	**Future scope**	26
8	**References**	27

LIST OF FIGURES

Fig 2.1: Vehicle suspension …….5

Fig 2.2: Power generating shock absorber ……..6

Fig 2.3: Working of regenerative shock absorber …..……7

Fig 2.4: Lead screw …….8

Fig 2.5: Spur gear…….8

Fig 2.6: Rectifier …….…….....9

Fig 2.7: Capacitor …….....10

Fig 2.8: LED bulb ……..……..11

Fig 2.9: Rotary generator……..…12

Fig 2.10: Suspension spring…….……12

Fig 2.11: Housing frame……..…….....13

Fig 2.12: Energy storage system……....13

Fig 2.13: Mounting components……14

Fig 3.1: Material selection…….15

Fig 3.2: Frame fabrication…….16

Fig 3.3: Actuator installation…….17

Fig 3.4: Electrical components……..18

Fig 3.5: Testing and calibration…….19

Fig 3.6: Final assembly isometric view…….20

CHAPTER-1

1.1 INTRODUCTION

The increasing global emphasis on energy efficiency and sustainable practices has catalysed innovations aimed at harvesting untapped energy sources. One such promising avenue lies in regenerative technologies, where energy that would otherwise be wasted is captured and reused. The "Regenerative Shock Absorber using Rack and Pinion Mechanism" project is a step toward integrating energy recovery solutions into everyday systems, particularly within the automotive sector.

Shock absorbers, a critical component of vehicle suspension systems, are designed to dissipate kinetic energy generated during motion, ensuring comfort and stability. Traditionally, this energy is lost as heat. However, by employing a regenerative mechanism, this otherwise wasted energy can be converted into useful electrical power. The rack and pinion mechanism serves as the core of this innovation, transforming the linear motion of the suspension system into rotational motion, which is subsequently harnessed to generate electricity.

This project aims to explore the design, functionality, and potential applications of such a system. By integrating mechanical ingenuity with energy recovery principles, the regenerative shock absorber could contribute significantly to improving vehicle energy efficiency. Furthermore, the study delves into the practical feasibility, expected performance, and possible limitations of implementing this technology in modern vehicles, paving the way for advancements in sustainable engineering practices.

In addition to its environmental benefits, the system holds economic potential by reducing the reliance on external energy sources and enhancing the overall efficiency of vehicles.

In the past decade, regenerative braking systems have become increasingly popular, recovering energy that would otherwise be lost through braking.

However, another energy recovery mechanism that is still in the research stages is regenerative suspension systems.

This technology has the ability to continuously recover a vehicle's vibration less energy dissipation that occurs due to road irregularities, vehicle acceleration, and braking, and use the energy to reduce fuel consumption.

Consumption; however, only 10%-16% fuel energy in the vehicles is utilized for driving to

The function of vehicle suspension system is to support the weight of vehicle body, to isolate the vehicle chassis from road disturbances, and to enable the wheels to hold the road surface.

Two chief elements in suspension are spring and damper.

Conventionally, damper is designed to dissipate vibration energy into heat to attenuate the vibration which is transmitted from road excitation.

However, the dissipated heat is from fuel or electrical power. It is a pity that so much energy is wasted.

Green manufacturing, also called environmentally conscious manufacturing, is one of the most popular topics nowadays. The future of green manufacturing technology is foreseeable, especially on vehicle industry.

Since the suspension is an important source of energy dissipation, it is feasible to harvest its vibration energy and convert into regenerative energy to improve the vehicle fuel efficiency. Therefore, so called regenerative suspensions arise as the times require

The project also emphasizes adaptability, ensuring the mechanism can be seamlessly incorporated into existing vehicle designs without compromising suspension performance or durability. Through rigorous analysis and testing, this initiative seeks to address challenges such as efficiency optimization, durability under varying conditions, and cost-effectiveness, offering a comprehensive perspective on the future of regenerative automotive technologies.

Looking ahead, the future of regenerative shock absorbers lies in advancing materials and technologies to further enhance energy conversion efficiency. Trends such as integrating smart sensors and IoT capabilities could enable real-time monitoring and optimization, improving system performance under diverse conditions.

Moreover, as electric and hybrid vehicles become more prevalent, regenerative shock absorbers can complement existing energy recovery systems, like regenerative braking, to maximize energy savings. Research into lightweight, durable materials and compact designs will also support broader adoption in a variety of vehicles, from passenger cars to heavy-duty trucks. This innovation represents a pivotal step toward creating smarter, more sustainable transportation systems for the future.

CHAPTER-2

2.1 SHOCK ABSORBER

A shock absorber is a mechanical device designed to smooth out or damp shock impulse, and convert kinetic energy to another form of energy (usually thermal energy, which can be easily dissipated). It is a type of dashpot.

A shock absorber is a device which convert mechanical energy into thermal energy.

The energy transformation occurs as the shock absorbers fluid medium is forced through orifice at high velocity.

Pneumatic and hydraulic shock absorbers are used in conjunction with cushions and springs.

An automobile shock absorber contains spring-loaded check valves and orifices to control the flow of oils through an internal piston.one design consideration, when designing or choosing a shock absorber, is where that energy will go.

In most shock absorbers, energy is converted to heat inside the viscous fluid. In hydraulic cylinders, the hydraulic fluid heats up, while in air cylinders, the hot air is usually exhausted to the atmosphere.

In other types of shock absorbers, such as electromagnetic types, the dissipated energy can be stored and used later. In general terms, shock absorbers help cushion vehicles on uneven roads.

The Power-Generating Shock Absorber (PGSA) converts this kinetic energy into electricity instead of heat through a Linear Motion Electromagnetic System (LMES). The LMES uses a dense permanent magnet stack embedded in the main piston, a switchable series of stator coil windings, a rectifier, and an electronic control system to manage the varying electrical output and dampening load. The bottom shaft of the PGSA mounts to the moving suspension member and forces the magnet stack to reciprocate within the annular array of stator windings, producing alternating current electricity. That electricity is then converted into direct current through a full-wave rectifier and stored in the vehicle's batteries. The electricity generated by each PGSA can then be combined with electricity from other power generation systems and stored in the vehicle's batteries to increase battery life. In non-electric vehicles the electricity can be used to power accessories such as air conditioning. Several different systems have been developed recently, though they are still in stages of development and not installed on production vehicles.

The mechatronic system we have chosen to model is Power-Generating Shock Absorber (PGSA) acting on an automotive chassis. The shock absorber will be used in conjunction with a spring to simulate one of the four suspension system of an automobile. When designing an automotive suspension system, the key is to balance the ride of the automobile.

More specifically, the suspension is meant to absorb the effects of an uneven driving surface and tilt/sway of the car.

However, excess energy loss occurs due to resistance in the damper fluid and compression of the spring. The PGSA converts kinetic energy into electricity through the use of a Linear Motion Electromagnetic System (LMES). As the absorber consists of a damper with permanent magnet stack that slide in and out of stator windings connected to two sliding blocks inside the damper casing.

Our model of a PGSA uses a modified EMF. The EMF was converted for translational motion and installed opposite of the traditional application. A power sensor with an integrator measures the energy generated by the reverse linear motor. The model is the strut-tower setup, so a spring was included to complete the model.

The Power-Generating Shock Absorber (PGSA) converts this kinetic energy into electricity instead of heat through the use of a Linear Motion Electromagnetic System (LMES). The LMES uses a dense permanent magnet stack embedded in the main piston, a switchable series of stator coil windings, a rectifier, and an electronic control system to manage the varying electrical output and dampening load.

The bottom shaft of the PGSA mounts to the moving suspension member and forces the magnet stack to reciprocate within the annular array of stator windings, producing alternating current electricity. That electricity is then converted into direct current through a full-wave rectifier and stored in the vehicle's batteries.

The electricity generated by each PGSA can then be combined with electricity from other power generation systems (e.g. regenerative braking) and stored in the vehicle's batteries. In turn, the electrical power can be used to recharge batteries or other efficient energy storage devices (e.g., flywheels) rather than be dissipated including four tires and entire suspension.

To accurately test our PGSA, we utilize four of the shock absorbers in a model of a traditional four-wheeled car. The electrical energy generated is the sum of the output from all four PGSAs. The ride quality in the car is characterized by the maximum vertical acceleration, taken by an accelerometer. The cost of the PGSA is calculated outside of Dipole in an Excel Spreadsheet. These objectives were analysed in Model Centre.

2.2 VEHICLE SUSPENSION

In a vehicle, shock absorbers reduce the effect of travelling over rough ground, leading to improved ride quality and vehicle handling. While shock absorbers serve the purpose of limiting excessive suspension movement, their intended sole purpose is to damp spring oscillations.

Shock absorbers use valving of oil and gasses to absorb excess energy from the springs. Spring rates are chosen by the manufacturer based on the weight of the vehicle, loaded and unloaded. Some people use shocks to modify spring rates but this is not the correct use. Along with hysteresis in the tire itself, they damp the energy stored in the motion of the unspring weight up and down.

Effective wheel bounce damping may require tuning shocks to an optimal resistance. Spring-based shock absorbers commonly use coil springs or leaf springs, though torsion bars are used in torsional shocks as well. Ideal springs alone, however, are not shock absorbers, as springs only store and do not dissipate or absorb energy. Vehicles typically employ both hydraulic shock absorbers and springs or torsion bars. In this combination, "shock absorber" refers specifically to the hydraulic piston that absorbs and dissipates vibration.

Fig 2.1: Vehicle suspension

2.3 POWER GENERATING SHOCK ABSORBER

The Power-Generating Shock Absorber (PGSA) converts this kinetic energy into electricity instead of heat through a Linear Motion Electromagnetic System (LMES). The LMES uses a dense permanent magnet stack embedded in the main piston, a switchable series of stator coil windings, a rectifier, and an electronic control system to manage the varying electrical output and dampening load. The bottom shaft of the PGSA mounts to the moving suspension member and forces the magnet stack to reciprocate within the annular array of stator windings, producing alternating current electricity. That electricity is then converted into direct current through a full-wave rectifier and stored in the vehicle's batteries. The electricity generated by each PGSA can then be combined with electricity from other power generation systems and stored in the vehicle's batteries to increase battery life.

In non-electric vehicles the electricity can be used to power accessories such as air conditioning. Several different systems have been developed recently, though they are still in stages of development and not installed on production vehicles.

Power Generating Shock Absorber Description The mechatronic system we have chosen to model is Power-Generating Shock Absorber (PGSA) acting on an automotive chassis. The shock absorber will be used in conjunction with a spring to simulate one of the four suspension system of an automobile. When designing an automotive suspension system the key is to balance the ride of the automobile. More specifically, the suspension is meant to absorb the effects of an uneven driving surface and tilt/sway of the car.

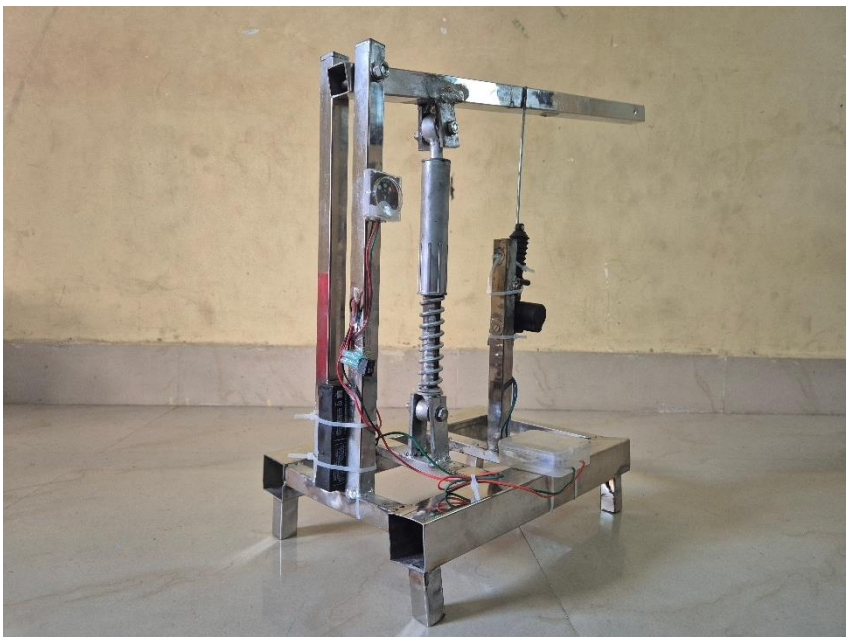

Fig 2.2: Power generating shock absorber

2.4 WORKING OF POWER GENERATING USING SHOCK ABSORBER

The Power-Generating Shock Absorber (PGSA) converts this kinetic energy into electricity instead of heat through the use of a Linear Motion Electromagnetic System (LMES). Shock absorbers are installed between chassis and wheels to suppress the vibration, mainly induced by road roughness, to ensure ride comfort and road handling. Plunger in shock absorber is the source of linear motion which is used to transmit the oscillatory motion to the actuator consisting of a lead screw connected the spur gear that drives the motor gear. The linear motion is converted into the rotary motion. Where the rotary motion is used to generate electricity in this mechanism. Then the electric motor converts the magnetic effect into electricity which is to be stored in battery. That electricity is then converted into direct current through a full wave rectifier and stored in the vehicle's batteries.

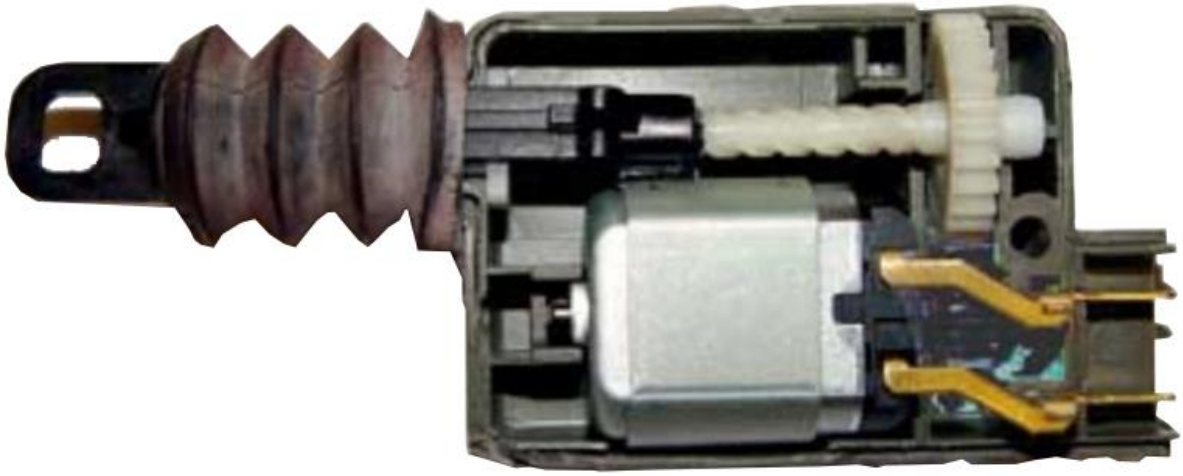

Fig 2.3: Working of Regenerative shock absorber

2.5 COMPONENTS

2.5.1 Parts of actuator

A regenerative shock absorber using a rack-and-pinion mechanism is an innovative design that not only dampens vibrations but also converts kinetic energy into electrical energy for storage or reuse. Below is a detailed description of each component:

2.5.1. Lead Screw

Description: A linear threaded bar that is connected to the suspension of the vehicle. It moves vertically in response to road-induced vibrations or oscillations.

Function: Converts the vertical motion of the suspension into linear motion, which drives the spur gear.

Material: Typically made of hardened steel or other durable materials to withstand constant motion and load.

2.5.2. Spur Gear

Description: A small, circular gear that is compounded with the teeth of Lead Screw.

Function: Converts the linear motion of the suspension system into rotational motion. This rotational motion is critical for driving the energy conversion mechanism.

Material: Often made of high-strength steel or alloy for durability and precision.

Fig 2.4: Lead Screw

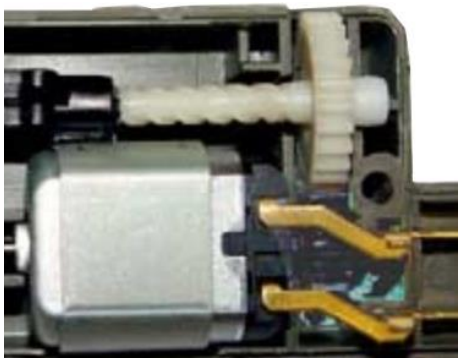

Fig 2.5: Spur Gear

2.5.2 Rectifier

A rectifier is an electrical device that converts alternating current (AC) to direct current (DC). This process is called rectification and is essential in power supply systems where DC is required, such as battery charging, electronic circuits, and power transmission.

2.6.1.Types of Rectifiers:

1. **Half-Wave Rectifier**
    - Uses a single diode to convert AC to DC.
    - Conducts only during the positive half-cycle of AC.
    - Simple design but low efficiency (around 40%).

2. **Full-Wave Rectifier**
    - Uses two or four diodes to convert both halves of AC to DC.
    - More efficient (around 80%) than half-wave rectifiers.
    - Two types:
        - Center-Tap Rectifier: Uses a center-tap transformer and two diodes.
        - Bridge Rectifier: Uses four diodes and does not require a center-tap transformer.

Working of a Bridge Rectifier:

- Four diodes are arranged in a bridge configuration.
- During the positive half-cycle, two diodes conduct and allow current flow.
- During the negative half-cycle, the other two diodes conduct, maintaining the same polarity.
- This results in continuous DC output.

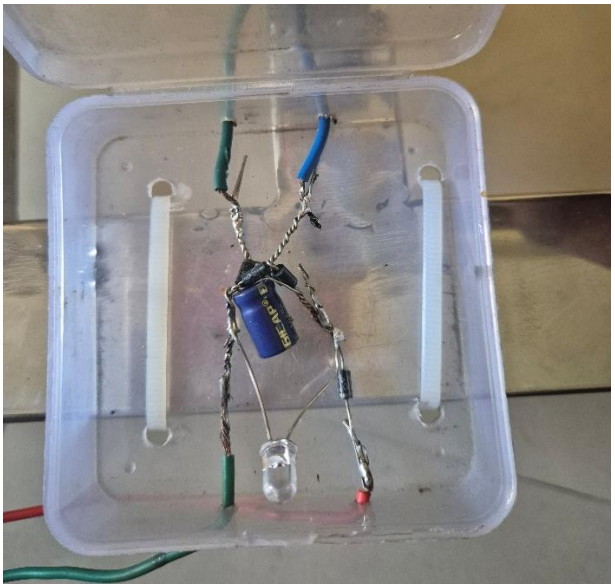

Fig 2.6: Rectifier

Applications of Rectifiers:

- Power supply circuits for electronic devices.

- Battery charging systems.
- DC motor drives.
- Signal demodulation in communication systems.

- **2.5.3 Capacitor**

  2.7.1. A **capacitor** is a passive electronic component that **stores and releases electrical energy** in a circuit. It consists of **two conductive plates separated by an insulating material (dielectric).**

  Basic Working Principle:
- When voltage is applied, **electric charge accumulates** on the plates, creating an electric field.
- When the supply is removed, the capacitor **releases stored energy** back into the circuit.
- The ability of a capacitor to store charge is called **capacitance (C)**, measured in **farads (F)**.

Fig 2.7: Capacitor

Types of Capacitors:

1. **Electrolytic Capacitors** – Used in power supplies and filtering circuits (high capacitance values).
2. **Ceramic Capacitors** – Used in high-frequency applications (small, stable, and non-polarized).
3. **Film Capacitors** – Reliable and used in AC circuits (good stability and low loss).
4. **Tantalum Capacitors** – Small and efficient for low-voltage applications.
5. **Supercapacitors** – Store a large amount of energy, used in backup power applications.

Applications of Capacitors:
- **Energy storage** in power supplies.
- **Smoothing rectified voltage** in rectifiers.
- **Filtering signals** in audio and communication circuits.
- **Timing circuits** in oscillators and microcontrollers.

2.5.4 LED bulb

Description: A small LED bulb is an energy-efficient, long-lasting lighting device that uses semiconductor technology to emit bright light with minimal power consumption.

Function: The function of an LED bulb is to convert electrical energy into visible light through the process of electroluminescence, providing efficient and long-lasting illumination

Material: LED lights typically use the following materials:

1. Semiconductor Materials: Commonly gallium arsenide (GaAs), gallium phosphide (GaP), or gallium nitride (GaN).
2. Substrate: Sapphire, silicon carbide, or silicon.
3. Lens: Epoxy resin or plastic to cover and protect the LED chip.
4. Phosphor Coating: Often used in white LEDs to convert blue light to white light.
5. Encapsulation: Transparent materials like silicone or epoxy to encase the semiconductor and improve light emission**.**

Fig 2.8: LED bulb

2.5.5 Rotary Generator

Description: A dynamo or motor-like device coupled to the gear.

Function: Converts the rotational motion into electrical energy. The generated electricity can be stored in a battery or used directly to power electronic components.

Material: Contains coils, magnets, and other electrical components designed for efficient energy conversion.

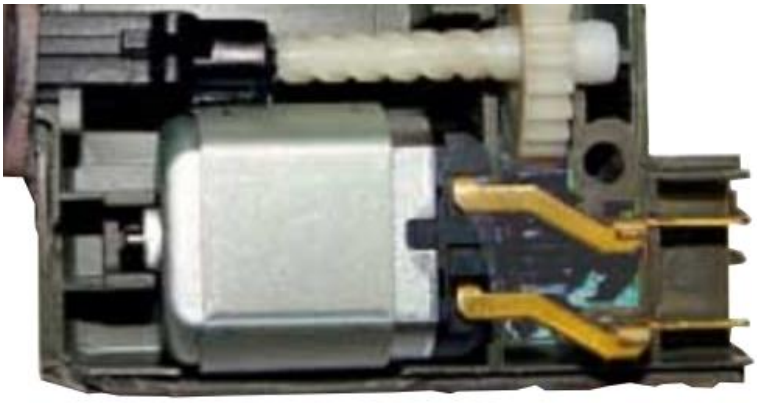

Fig 2.9: Rotary generator

2.5.6 Suspension Spring

Description: A standard spring component that works in tandem with the rack-and-pinion system.

Function: Absorbs and dissipates kinetic energy to reduce the impact on the vehicle's chassis while working with the shock absorber mechanism.

Material: Typically made of high-tensile steel.

Fig 2.10: suspension spring

2.5.7 Housing

Description: An outer casing that encases the rack, pinion, and other internal components.

Function: Protects the internal mechanism from dirt, water, and other external elements. It also provides structural support.

Material: Made of durable materials like aluminium or steel to ensure longevity and lightweight construction.

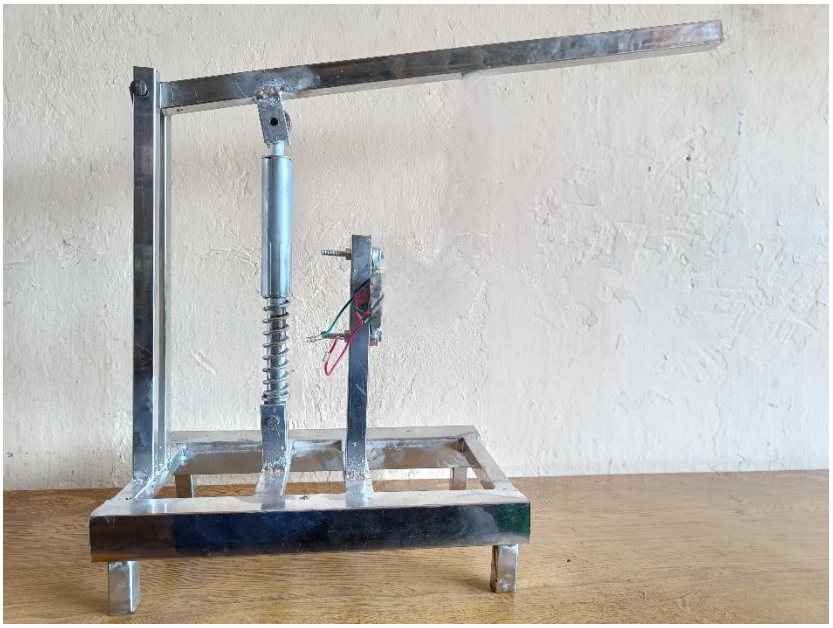

Fig 2.11: Housing frame

2.5.8 Energy Storage System

Description: A battery or capacitor system connected to the rotary generator.

Function: Stores the electrical energy generated by the rotary generator for later use, such as powering auxiliary systems or feeding back into the vehicle's powertrain.

Material: Depends on the storage technology; lithium-ion batteries are common.

Fig 2.12: Energy storage system

2.5.9 Control Electronics

Description: Circuit boards and microcontrollers that manage the energy flow.

Function: Regulates the voltage and current generated, ensuring efficient energy transfer to the storage system and monitoring system performance.

Material: Composed of semiconductors, resistors, and capacitors for precise control.

2.5.10 Mounting Brackets

Description: Components used to attach the regenerative shock absorber to the vehicle chassis.

Function: Ensures secure installation and optimal alignment for the system to operate effectively.

Material: Made of steel or aluminium for strength and resistance to wear.

Fig 2.13: Mounting Components

2.5.11 Shock Absorbing Fluid or System

Description: May include hydraulic fluid or other damping materials depending on the design.

Function: Provides traditional shock absorption to complement the energy regeneration process, ensuring ride comfort.

Material: High-viscosity synthetic or mineral oils.

This system represents a blend of mechanical and electrical engineering, designed to improve energy efficiency and sustainability in modern vehicles.

CHAPTER-3

3.1 FABRICATION PROCESS

Process involved in fabrication of power generating shock absorber

- Design and material selection
- Frame fabrication
- Mounting the shock absorber
- Actuator installation
- Electrical connections
- Testing and calibration
- Final assembly check

3.1.1 Design and Material Selection

Theory: A regenerative shock absorber converts the linear motion from vibrations into rotational motion using a rack and pinion mechanism. This rotational energy is then fed into a generator, producing electrical energy.

Materials Required:

Shock Absorber: A standard suspension shock absorber with a piston and spring.

Metal Frame: Steel rods, pipes, or plates for structural support.

Actuator: Coverts mechanical energy into electrical energy.

Fasteners: Nuts, bolts, and clamps.

Wires and Load: Electrical wires, multimeter, and resistive load like a bulb or battery.

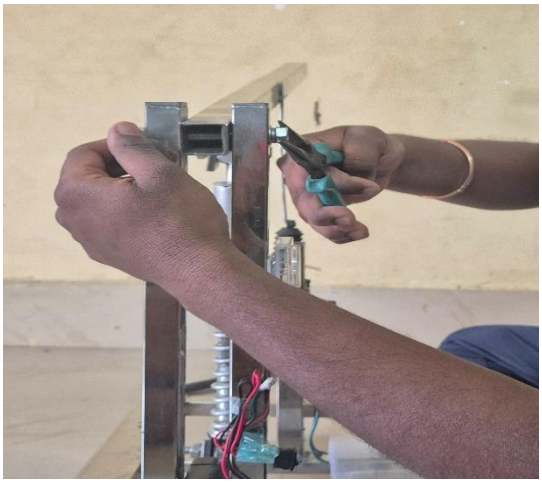

Fig 3.1: Material selection

3.1.2 Frame Fabrication

Purpose: To provide a rigid structure to mount the components

Steps:

Design the Frame:

Sketch a layout for the frame to house the shock absorber, rack, pinion, and generator.

Ensure the frame can withstand the mechanical forces during compression and expansion.

Cutting the Frame Materials:

Use mild steel rods or rectangular pipes.

Cut the materials into required dimensions using cutting tools like grinders or saws.

Welding or Bolting:

Assemble the cut pieces into a rectangular or L-shaped frame.

Weld the joints for rigidity or use bolts and nuts for modular construction.

Surface Finish:

Clean the welded surfaces and apply anti-corrosive paint to ensure durability.

Fig 3.2: Frame fabrication

3.1.3 Mounting the Shock Absorber

Theory: The shock absorber will provide linear motion when compressed, which is essential to drive the rack.

Steps:

Positioning the Shock Absorber:

Fix the bottom of the shock absorber securely to the base of the frame using clamps or bolts.

Ensure the shock absorber stands vertically.

Testing Movement:

Apply downward force to ensure the piston rod moves smoothly under compression and returns under spring force.

Alignment:Align the piston rod so it moves perfectly vertical to avoid misalignment during motion conversion.

3.1.4 Actuator Installation

Theory: The lead screw (linear gear) converts the shock absorbers linear motion into rotational motion by engaging with the pin gear (circular gear).

Steps:

Actuator Attachment:

Fix the actuator to the frame rod of the shock absorber.

Use clamps or weld it securely, ensuring the rack remains vertical and aligned with the spur gear.

Ensure the shaft reciprocate smoothly when the absorber moves.

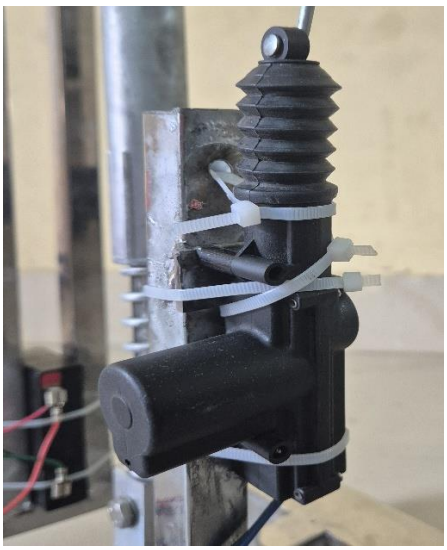

Fig 3.3: Actuator installation

3.1.5 Electrical Connections

Theory: The actuator will produce voltage proportional to the speed of rotation.

Steps:

Wiring the Generator:

Connect the generator's output terminals using electrical wires.

Load Connection:

Attach the generator output to a resistive load like a bulb, battery, or testing equipment (multimeter).

Circuit Safety:

Use connectors and insulate wires to avoid short circuits.

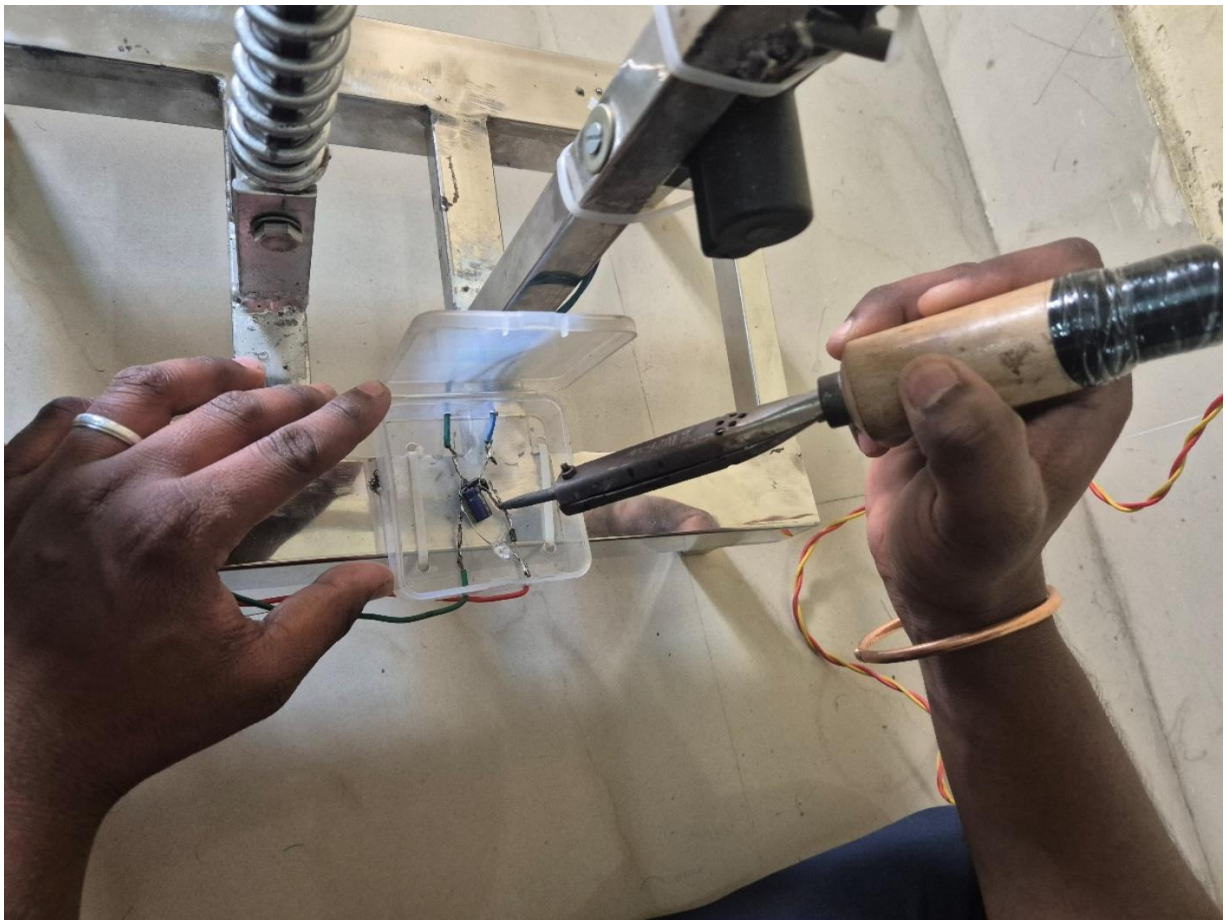

Fig 3.4: Electrical connections

3.1.6 Testing and Calibration

Steps:

Manual Testing:

Apply a vertical load to the shock absorber, causing the piston rod to move.

Generator Output:

Use a multimeter to measure the generator's voltage and current output.

Check for smooth and consistent power generation.

Load Testing:

Connect a small bulb or charge a battery to demonstrate energy generation.

Increase or decrease the applied force to observe changes in power output.

Alignment Adjustments:

If the rack and pinion misalign, adjust their positions to ensure smooth engagement.

Friction Reduction:

Fig 3.5: Testing and calibration

3.1.7 Final Assembly Check

Ensure all components (shock absorber, rack, pinion, generator, and frame) are securely mounted.

Recheck electrical wiring for safety and efficiency.

Verify smooth motion of the rack and pinion system under load.

Working Principle Recap

Linear Motion: Vertical force compresses the shock absorber, moving the rack linearly.

Rotational Motion: The linear motion of the piston drives the actuator to rotate the gear.

Power Generation: The rotating gear powers the generator, converting mechanical energy into electrical energy.

Conclusion: The regenerative shock absorber using door lock actuator mechanism is now fabricated and operational. It can recover energy from vibrations and shocks, which can be stored or used directly for various small applications like lighting or charging batteries. Lubricate all moving parts, such as the pinion, shaft, and bearings, to enhance efficiency.

Fig 3.6: Final assembly isometric view

CHAPTER-4

4.1 ADVANTAGES

1. **Energy Efficiency**: By converting otherwise wasted kinetic energy into electricity, the system contributes to improved energy efficiency in vehicles, reducing reliance on external power sources.
2. **Environmental Impact**: The regenerative shock absorber supports sustainable practices by decreasing energy wastage, aligning with global efforts to combat climate change.
3. **Economic Benefits**: The system has the potential to lower operational costs for vehicles by supplementing energy needs, thereby reducing fuel or electricity consumption.
4. **Design Versatility**: The rack and pinion mechanism is a proven mechanical concept that can be adapted to various vehicle types, from passenger cars to heavy-duty trucks.
5. **Integration Potential**: The system can complement existing regenerative technologies, such as regenerative braking in hybrid and electric vehicles, enhancing overall energy recovery.
6. **Innovation in Mobility**: Projects like this pave the way for the development of smarter vehicles that utilize cutting-edge technologies for better performance and sustainability.
7. **Corrosion Resistance**: Stainless steel is highly resistant to corrosion, making it an ideal material for automotive applications that are frequently exposed to moisture, dirt, and varying weather conditions. This enhances the longevity and reliability of the housing framework.
8. **Durability and Strength**: Known for its high tensile strength, stainless steel can endure significant mechanical stress and vibrations, ensuring that the shock absorber's housing remains intact under challenging operational conditions.
9. **Low Maintenance**: The material's resistance to wear and corrosion reduces the need for frequent repairs and replacements, contributing to lower maintenance costs over the lifespan of the vehicle.
10. **Recyclability**: Stainless steel is 100% recyclable, aligning with the project's sustainability goals by reducing environmental impact and promoting the use of eco-friendly materials.
11. **Aesthetic Appeal**: The sleek and polished finish of stainless steel adds a premium appearance to the shock absorber, which could be an attractive feature for high-end automotive markets.
12. **Heat Resistance**: Its ability to withstand high temperatures ensures stable performance in diverse operating environments, especially during prolonged or intensive use.
13. **Compatibility**: Stainless steel can be easily fabricated and integrated with other components, simplifying the manufacturing and assembly processes.

4.2 DISADVANTAGES

1. **Complexity in Design**: Incorporating a regenerative system into existing vehicle designs without affecting suspension performance can be challenging.
2. **Cost Concerns**: The initial development and manufacturing of the system may be expensive, potentially limiting widespread adoption.
3. **Durability Issues**: The mechanism must withstand continuous operation under diverse road conditions, posing challenges to material selection and design.
4. **Energy Conversion Efficiency**: Achieving high efficiency in energy conversion may require advanced materials and technologies, which can add complexity.
5. **Weight Considerations**: Adding regenerative components to vehicles may increase weight, potentially affecting fuel efficiency and performance.
6. **Maintenance Challenges**: The system's added components could lead to increased maintenance requirements and costs over time.
7. **High Cost**: Stainless steel is relatively expensive compared to alternative materials such as aluminium or mild steel, which could increase the overall cost of manufacturing the regenerative shock absorber.
8. **Weight Factor**: The higher density of stainless steel contributes to increased weight, which might negatively impact the overall fuel efficiency of the vehicle. This is a critical consideration for automotive applications prioritizing lightweight designs.
9. **Machining Challenges**: While stainless steel is durable, its hardness can make machining and fabrication more time-consuming and expensive. Specialized tools and techniques may be required, increasing production costs.
10. **Thermal Conductivity**: Stainless steel has lower thermal conductivity compared to other metals like aluminium, which could result in localized heat buildup under specific conditions, potentially affecting performance.
11. **Initial Investment**: The higher upfront costs for material procurement and processing might deter widespread adoption, especially in budget-sensitive automotive markets.
12. **Sustainability Concerns in Mining**: Although recyclable, the extraction and processing of stainless steel involve significant environmental impact, which could offset some of its sustainability benefits.
13. **Potential for Galling**: Under certain conditions, stainless steel components can suffer from galling, a form of wear caused by adhesion between sliding surfaces. This could necessitate additional treatments or coatings to mitigate the issue.

4.3 APPLICATIONS

This technology can be applied to any type of vehicle that employs movable suspension technology and uses electricity in some form as its fuel. It is successfully tested on electric vehicles. The system performs best on heavy, off-road vehicles moving quickly over rough terrain, so the company is targeting military. applications. It also is sensible that having onboard power generation could be a real advantage in military situations where troops are moving in remote areas without readily available fuel sources. Conserving fuel in those scenarios, especially during combat, could be the difference between life and death.

What comes to mind quickly for non-military applications is the commercial trucking Industry. While they typically run trucks over roadways, their payloads of tens of thousands of pounds couple even with small, constant movements might generate a fair amount of electricity with shock absorber generators. To improve vehicle handling, the power controller uses information from accelerometers and other sensors to change the resistance from the generators, which stiffens or softens the suspension.

For example, if the sensors detect the car starting a turn, the power controller can increase the resistance from the shock absorbers on the outer wheels, improving covering, The shock absorbers are connected to a power management system that can interface with other sources of power, such as regenerative braking systems, thermoelectric devices that can convert waste heat into electricity.

4.4 ADVANCEMENTS

Recent advancements in regenerative shock absorber technology have focused on enhancing efficiency, adaptability, and integration with modern systems. One significant development is the incorporation of **advanced energy storage solutions** such as high-capacity batteries or supercapacitors, which can store energy more effectively for future use. This enables a smoother and more efficient transfer of energy from the shock absorber to the vehicle's power system.

Another promising trend is the integration of **smart sensors and IoT technologies**, allowing real-time monitoring and optimization of the energy recovery process. These sensors can adjust the system's performance based on road conditions, vehicle load, and driving behaviour, ensuring maximum energy recovery under varying circumstances.

In terms of materials, the use of **lightweight composites and alloys** has reduced the overall weight of the shock absorber system, minimizing its impact on vehicle performance while maintaining durability. Moreover, advancements in **magnetic and electromagnetic systems** have improved the efficiency of energy conversion, making the mechanism more viable for widespread adoption.

CHAPTER-5

5.1 CALCULATION

Electrical Calculation

Voltage Generated (V) = 1.8 volt

Current Generated (I) = 2.0 amp

As Electrical Power (P) = V x I = 1.8 x 2 = 3.6 Watts

To Calculate Charging Time for 6 Volt 2Amp Battery

Charging time =battery current (Ah) / current generated (A)

= 3.3 (Ah) /2 (A)

= 1.65 hours

But it was noted that during charging 40% get loss

= 4.5 x 40 /100

=1.8 Ah

Charging time = 3.3 +1.8 /2

= 4.2 hours

To charge a 6-volt battery with this suspension including 40% loss we can charge the battery with 4.2 hours.

CHAPTER-6

CONCLUSION

- The project reviews the background of research on regenerative shock absorber, and studies the vibration energy recovery potential of automobile vehicles. Then, the structures of the regenerative shock absorber are summarized and classified, which states the electric regenerative shock absorber can provide better performance and regenerated energy in terms of its specific design. Finally, some other approaches of modelling and fabrication are introduced for follow-up work.

- This project also has certain limitations such as it has certain space constraints. The system should acquire minimum space and should not hamper the damping capacity of the suspension system. The system was designed to generate 6 volts of output but the actual system generated 4 volts of output which was less than the expected due to losses such as frictional loss and alignment loss

- The system which was designed in this project is suitable for the vehicles where sufficient space is available for accommodating the parallel suspension and regenerative system. We are in the era where the companies are keen on making system compact by using advanced direct conversion devices. Here in our system, we can place the regenerative system in the suspension. This will lead to a compact design on suspension. As a future scope we can completely replace the conventional damper by mechanical generator and the vibration damping can be provided by the resistance to motion by the generator. This will lead to total vibration energy conversion and better results.

CHAPTER-7

FUTURE SCOPE

The regenerative suspension is moderately another improvement and has a high degree for future upgrade. Numerous researchers have put forth the attempt to begin the improvement of regenerative suspension and bring this into consideration for various potential analysts. Being at its condition of early stages, regenerative suspension is certainly a sprouting point of research in the car business. It tends to be introduced in commercial vehicles, trucks and multi utility vehicles for invigorating the helper electrical part

• Such suspension systems can be designed for heavy vehicles, thus increasing input torque to the regenerative system, and ultimately output of the generator.

• More suitable and compact mechanisms to enhance efficiency of the system.

• Various manufacturing companies can take up an initiative to implement these power humps on a large scale in the market.

• This has a huge scope everywhere provided the resources are channelled well.

CHAPTER-8

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